Federated Learning-Based Intrusion Detection System for Privacy-Preserving IoT Networks

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1 Introduction

The Internet of Things (IoT) composed of interconnected physical devices equipped with sensing, communication, and computing capabilities allowing them to exchange of information via internet-based networks [1, 2]. The continuous expansion of these devices has impacted numerous aspects of daily life, from smart homes, smart cities and healthcare to industrial systems. By connecting these devices and enabling real-time data exchange, IoT technologies improve efficiency, speed, and decision making across various processes. In addition to these benefits, the widespread adoption of IoT devices also introduced significant security and privacy concerns, making the ecosystem a prime target for cyber-attacks [3].

To safeguard IoT networks from such threats, an intrusion detection system (IDS) has been developed to monitor network traffic and detect malicious activities. To extract meaningful insights from the data and support detection systems, machine learning (ML) techniques have been widely adopted. Cyber attacks detection is one of the key intelligent applications enabled by these techniques [4]. However, traditional centralized IDSs frequently face challenges in handling these tasks effectively. Collecting and processing vast volumes of data in a single central location can compromise privacy, generate substantial communication overhead, and can create a single point of failure [5].

A practical example of these challenges can be observed in smart home environments, where interconnected IoT devices such as smart cameras, voice assistants, motion sensors, and home automation controllers continuously collect and exchange sensitive data. These devices often handle personal information related to daily routines, occupancy patterns, and security status. In traditional centralized intrusion detection systems, this data is transmitted and aggregated on a central server for analysis. Although centralization simplifies model training, it introduces significant disadvantages, including increased privacy risks due to raw data exposure, communication overhead, and scalability bottlenecks. [6].

To address the limitations of centralized approaches, Federated Learning (FL) offers a decentralized training framework well suited for IoT environments. In this paradigm, each participating device independently trains machine learning or deep learning models using locally stored data. Instead of transmitting raw datasets, devices exchange only model parameters or gradient updates to a coordinating server, which then aggregates these model updates to improve a shared global model. This approach preserves data privacy while enabling collaborative learning across multiple devices [5, 7].

Liu et al. (2026), highlighted that despite the advantages FL offers for IoT IDS, several critical challenges persist with its deployment. Frequent model updates can strain limited network bandwidth, and algorithms must handle diverse device capabilities and data distributions without compromising accuracy. FL is also vulnerable to threats such as model poisoning and adversarial attacks, which require robust security measures. Additionally, IoT devices limited computational resources, along with non-IID data make efficient, scalable, and reliable deployment challenging [8].

In smart home environments, various IoT devices are used for day to day tasks resulting in generating an enormous amount of sensitive data that are being exchanged between devices. If these systems are compromised, the impact extends beyond technical disruption, potentially affecting user privacy, personal safety, and overall trust in connected technologies. Effective resolution of these challenges

is essential for the implementation and deployment of FL-based IDS in IoT networks. This requires the development of lightweight optimized models that can operate efficiently on resource constrained devices with limited computational power, battery capacity, and bandwidth, while maintaining high detection accuracy. Overcoming these challenges will enable IoT devices to counter emerging cyber attacks while ensuring privacy, scalability, and reliability in diverse IoT environments.

2 Literature Review

The performance of the IDS is highly dependent on the datasets. Collecting and analyzing a realistic IoT dataset is crucial for this task. According to the literature, there are many benchmark IoT datasets available such as CIC-IoT-2023, NSL-KDD, N-BaIoT, IoT-23, MQTT-IoT-IDS2020, Ton_IoT, and X-IIoTID [9]. Pinto Neto et al. [2] published the CIC-IoT-2023 dataset which contains 33 attack types.

The study conducted in [10] utilized the CIC-IoT-2023 dataset and proposed an LSTM architecture to classify the attack types. The proposed model achieved an accuracy of 98.75 %. The study in [11] used the same dataset and proposed a transformer based IDS model. This model showed the superior performance over other state of the art model including DNN, RNN, CNN, and LSTM particularly in multi-class classification, achieved an accuracy of 99.40%.

FL has become a key approach for building IDS in IoT networks, providing an effective solution to issues of data privacy, security, and scalability [12, 13]. Given the distributed nature of IoT devices and the sensitive information they often handle, traditional centralized machine learning approaches for IDS are increasingly insufficient due to privacy concerns, computational resource limitations, and communication overhead [5, 14].

Several studies highlight the advantages of FL in this context of privacy preservation, scalability, communication costs reduction and resource efficiency. FL ensures that sensitive raw data from IoT devices remains on the local devices [5, 13]. This protects user privacy, by using differential privacy framework (DP) to add noise to updates, further obscuring individual data contributions but often introduces a trade-off with detection accuracy due to the added noise [13].

Traditional centralized IDSs often struggle with the scalability demands of large-scale IoT ecosystems. FL-based systems can accommodate a vast number of devices by distributing the computational burden, making them inherently more scalable [15]. By performing local training, FL reduces the need to transfer large volumes of raw data to a central server, thereby lowering communication costs and bandwidth requirements [14, 5]. FL can be optimized for resource-constrained IoT devices, making it feasible to deploy detection models directly on the edge [15].

The study in [16] utilized the benchmark MQTT dataset. It proposed a FL based CNN framework prioritizing optimal feature selection with chimp optimization and did a comparative analysis with traditional ML models such as decision tree, random forest, neural network, gradient boost and naïve bayes algorithms. The proposed model showed a high multi-class classification accuracy of 95.59% compared to the traditional ML models.

The study in [8] used Edge-IIoTset dataset and designed a framework that handles both IID and non-IID data distribution using CNN, DNN and Transformer models under Federated learning setup. It highlighted that the Transformer model outperformed the other two models(CNN, DNN) in both IID and non-IID settings.

The researchers in this study [17] proposed FL based TabTransformer model with an integeration of Proximal Policy Optimization(PPO) for optimaztion of classification detection. They evaluated the model on TON_IoT benchmark dataset and achieved an accuracy of 98.85%. Notably, It showed a robust performance in detecting the rare man-in-the-middle attacks, achieved an acuracy of 88.7%.

Recent studies have highlighted the potential of FL to enhance IDS with collaborative learning. The integration of deep learning techniques such as MLP, LSTM, CNN, DNN, RNN, and TabTransformer gave promising results on the benchmark IoT datasets. Among these the TabTransformer model showed strong results in terms of detection accuracy. Studies have emphasized the effectiveness of TabTransformer model framework for capturing the complex relationships in IoT traffic, making it a suitable choice for heterogeneous IoT environments [18, 19]. However, model optimization and optimal feature selection remain underexplored and moreover Lightweight models, which are computationally efficient and require less memory and energy, remain undiscovered in the context of FL based IDS.

The studies discussed in this section highlighted the need for optimized deep learning models that not only maintain high detection performance but are also efficient in resource constrained and heterogenous IoT devices. Although FL offers privacy through decentralization, model updates may still leak sensitive information [13]. Existing studies have not discussed the integration of diffrential privacy mechanism in the FL based TabTransformer framework, particularly in the IoT IDS. This highlights a research gap in developing a model that balances privacy preservation through differential privacy and accuracy.

3 Purpose and objectives

This research explores the development of a federated learning based IDS for IoT. We aim to design, implement, and evaluate an efficient, privacy-preserving, and scalable federated learning based IDS capable of accurately detecting cyber attacks in IoT networks.

To achieve this, we focus on the following objectives:

Design & Optimize a Federated based deep learning Architecture: Develop efficient federated learning based lightweight deep learning models and optimize for local training on resource constrained IoT devices while maintaining high detection accuracy and precision and minimizing computational cost and memory usage, and reducing communication overhead during federated model updates.

Integrate Privacy Mechanisms: Integrate the privacy mechanisms during model updates to enhance privacy and evaluate the trade-offs between privacy and detection performance.

Evaluate Scalability: Assess system performance as the number of IoT clients increases, focusing on detection accuracy and scalability.

Robustness to Zero-Day Attacks: Investigate the ability of the federated IDS to detect previously unseen (zero-day) attacks, ensuring the system generalizes effectively across diverse attack types and IoT scenarios.

4 Research Questions

Based on the identified research gaps, this study focuses on federated intrusion detection in IoT environments, with particular emphasis on smart home scenarios. To address these challenges, the following research questions are formulated:

RQ1: How can TabTransformer and lightweight deep learning models be designed and optimized to achieve effective intrusion detection while maintaining computational efficiency in federated IoT environments?

RQ2: How does the differential privacy technique affect the detection performance, and communication overhead under varying privacy budget, and what trade off arises?

RQ3: How does the federated IDS system scale with an increasing number of participating devices, in terms of detection performance, convergence speed, and communication overhead?

RQ4: How effective are the optimized federated models in detecting previously unseen (zero-day) attacks in IoT environments?

5 Research Methodology

5.1 Research Approach

This study adopts a Design Science Research (DSR) methodology combined with Experimental Research to develop and evaluate a Federated Learning based IDS for IoT networks.

Design Science Research methodology is used to systematically design and develop a Federated Learning-based IDS for IoT networks in smart home environments. DSR focuses on creating and evaluating a novel artifact that addresses a real-world problem. In this research, the problem of privacy risks and scalability limitations in traditional IDS solutions are identified and clearly defined. Based on these challenges, system objectives are established, emphasizing privacy preservation, detection accuracy, efficiency, scalability, and robustness against emerging and zero-day cyber threats.

Experimental research methodology is employed to empirically evaluate the performance of the models. The system is evaluated using standard performance metrics, including detection accuracy, precision, recall, F1-score, communication overhead, and training time. To assess scalability, the number of IoT clients is progressively increased, and the impact on detection performance and system efficiency is analyzed. Experimental approach ensures that the proposed system is validated quantitatively and its effectiveness under varying network scales is rigorously assessed.

The combination of Design Science and Experimental Research is appropriate because the study aims to build a novel technical artifact (FL-based IDS). This requires empirical validation of its effectiveness. Quantitative evaluation is conducted to assess accuracy, scalability and privacy preservation. This integrated approach ensures systematic system development and rigorous performance validation.

5.2 Dataset

The CICIoT2023 dataset is a large scale, real time benchmark designed to support the development of network intrusion detection systems for modern Internet of Things (IoT). It was created by researchers at the Canadian Institute for Cybersecurity (CIC) to address limitations in existing datasets, such as small network topologies and the lack of attacks executed specifically by IoT devices. They developed a dataset by executing 33 attacks in an IoT topology composed of 105 devices [2]. The dataset topology was designed to resemble a realistic smart home IoT environment. It includes a diverse range of interconnected devices such as smart home appliances, surveillance cameras, sensors, and microcontrollers. These devices are configured to communicate within a shared network, simulating how IoT products and services operate in real-world household settings [2].

The dataset comprising over 46 million samples and 34 unique attack types. These attacks are futher classified into seven categories, namely DDoS, DoS, Recon, Web-based, brute force, spoofing, and Mirai. All attacks are launched by compromised IoT devices against other IoT devices. The resulting dataset is available in two formats and contains 47 extracted features in the CSV version [2]. It is publicly available on the CIC Dataset website and has rapidly become a standard benchmark for evaluating the performance of IDS.

Following that, we will perform data preprocessing which involves data cleaning, categorical variable encoding, normalization and Feature extraction. This step ensures that the data is ready for training the model.

5.3 Model Design

A federated learning based deep learning models will be designed and optimized for efficient and privacy preserving intrusion detection for IoT environments.
This study will investigate two categories of models:
A TabTransformer (TTF) is a deep learning architecture designed for tabular data modeling using contextual embeddings [18, 19]. It uses the self-attention mechanism, originally popularized in natural language processing to capture complex, long range dependencies and interactions between different features in a dataset [19].

Lightweight Deep Learning Models:
In addition to TabTransformer, lightweight deep learning models such as Tabnet, 1D CNN, and MLP will be developed and optimized for deployment on resource-constrained IoT devices.

Privacy Integration: The differential Privacy mechanism will be integrated into the proposed framework to see how it affects the detection performance.

Scalability: To ensure the model scalability, it will be tested by simulating the multiple number of clients and analyzing the impact on the detection performance, model convergence and scalability.

Evaluation: The system will be evaluated using metrics such as accuracy, precision, recall, F1-score, and AUC-ROC.

Zero-Day Attack Evaluation: To evaluate the robustness and generalization capability of the proposed models, zero-day attack experiments will be conducted. Certain attack categories will be excluded during the training phase and later introduced during testing to simulate previously unseen attack scenarios. The performance of both TabTransformer and lightweight models will be assessed based on their ability to detect these unseen attacks using evaluation metrics.

6 Expected results

The federated learning-based IDS is expected to achieve better detection performance. Specifically, both the TabTransformer and lightweight deep learning models are expected to achieve high classification accuracy, precision, recall, F1-score, and AUC-ROC in detecting attacks, while maintaining low false positive rate.
With the integration of differential privacy mechanism, the system is expected to learn about the trade-off between privacy and detection accuracy.
The proposed federated learning framework is expected to maintain stable performance as the number of participating devices increase. Furthermore, the optimized federated models are expected to demonstrate the ability to detect previously unseen (zero-day) attacks,

7 Resource Specifications

Hardware Resources: CPU/GPU power and RAM are required to train and test the deep learning models. This project will use an AMD Ryzen 7 5700U processor with Radeon Graphics and 16 GB of RAM.

Software Resources: Software Resources: Python 3.11 will be used along with relevant deep learning libraries, including PyTorch (or TensorFlow), NumPy, Pandas, and SciPy. The Jupyter Notebook and the VS Code will be used for development and experimentation. Federated learning frameworks such as Flower or PySyft will be used to implement decentralized training. Data visualization tools such as Matplotlib, Seaborn will be used for result analysis.

Meeting Plan: The supervisors expertise in IoT networks, and machine learning will help in shaping the research. We plan to meet our supervisor biweekly or based on the availability to discuss,

progress and address technical challenges. The regular feedback will be sought to track progress, and ensure the final thesis maintains academic quality.

8 Risks

Table 1: Risk Assessment

Risk	Probability	Severity	Mitigation Strategy
Data Quality and Imbalance	Moderate	High	Data cleaning, remove redundant features, apply resampling techniques.
Computational Resource Constraints	Moderate	High	Implement lightweight deep learning architectures, apply quantization, and optimize training settings.

Data Quality and Imbalance Risk: Poor data quality may reduce model accuracy and reliability. This will be addressed through preprocessing, feature selection, and balancing techniques such as oversampling or undersampling.

Computational Resource Constraints: Federated and deep learning models may require significant computational power, potentially slowing experimentation. To mitigate this, lightweight architectures with reduced parameters and quantization techniques will be implemented.

9 Time plan

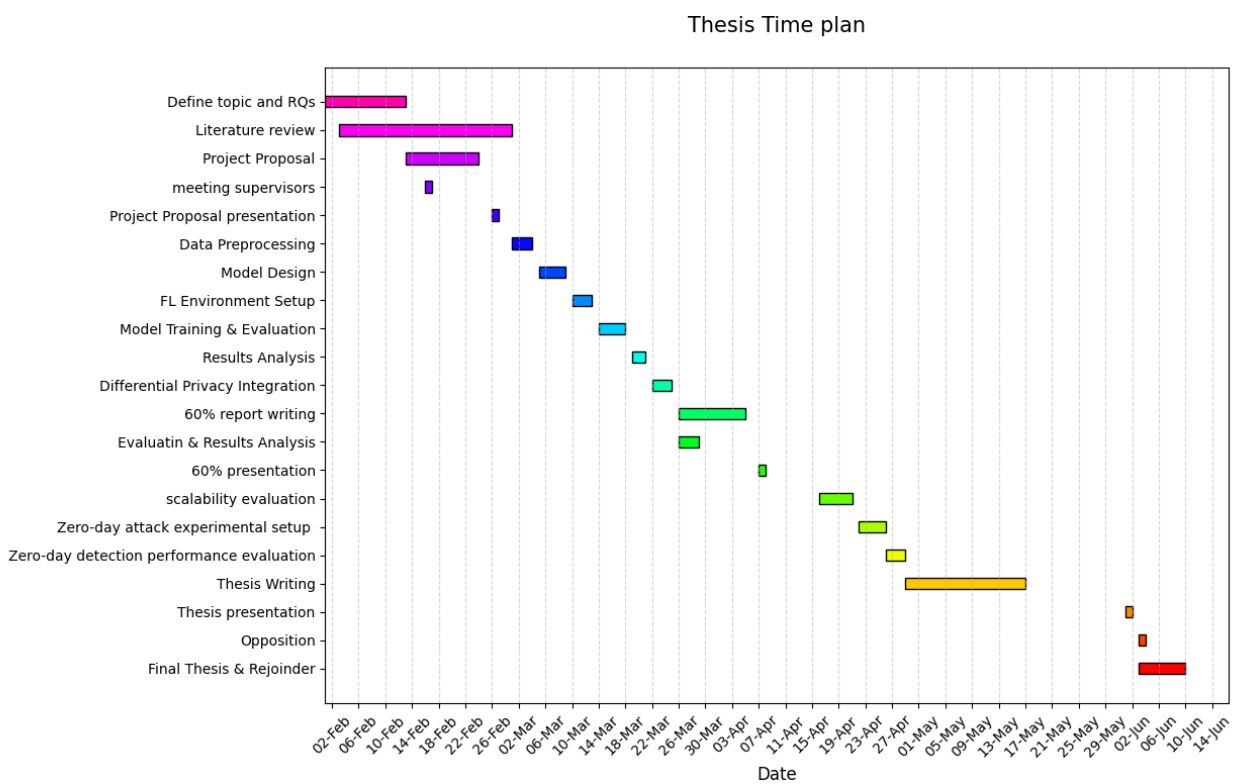

Figure 1: Time plan

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